Run report: lit2026 W8 diagnostics, run-20261011T2017Z

Started 2026-10-11 20:17 UTC. **Host** shared Apple M4 Pro, macOS (Darwin 25.6.0), load average 72–360 during the parts (`uptime` per round in every JSON). **Isolation** L0. **Harness** `examples/lit2026_w8_summation_models.rs` on branch `lit2026/w8-summation-models`; library byte-identical to program baseline `4b6e09b78` (`git diff 4b6e09b78 HEAD -- src Cargo.toml Cargo.lock` empty). This report describes what ran; interpretation is in the study README.

Parts, commands, binaries

Every part: `--rounds 5 --targets 8`, arms interleaved with the order rotating per round, seed `0x5EED`. Binaries are successive builds of the same example; SHA-256 of each and what changed:

file	binary (sha256 prefix)	env	status
`cert.json`	`420148c1` (re-issued by `da887b81`, byte-identical result)	—	complete
`lucas.json`	`420148c1`	default	complete (n = 9, 12, 15)
`lucas-l34.json`	`b42e8494`	default	complete (n = 9, 12); word ops equal `lucas.json`
`wu-first-binary-failed.json`	`420148c1`	default	failed: binary Edwards fit used 3 points that repeat ±P (bug, fixed)
`wu.json`	`da887b81`	default	complete
`eg-aborted-first-binary.stderr`	`420148c1`	default	stopped (rerun with ladder-only env)
`eg.json.partial`	`5216edd7`	`KIC_F4_MAX_DEGREE_ONLY=0`	9 of 10 cells; K1/F2^17 m=3 stopped after 2 h (SIGTERM)
`frob-aborted-first-binary.stderr`	`420148c1`	default	stopped: aux arm at 28 vars on the top-degree-only path, > 17 CPU-min in cell 1
`frob-aborted-w8bin3.*`	`5216edd7`	`=0`	stopped to add the engine flag; 3 cells recorded
`frob.json`	`99a4a31b`	`=0`	complete, MatrixF4
`frob-inherited.json`	`99a4a31b`	`=0`	complete, `--engine inherited`

Builds: `420148c1` first; `da887b81` adds the binary-Edwards fit against `X = x + sqrt(b)/x` (Wu part only); `5216edd7` adds per-cell checkpoints, the env record and the reduced `frob` cell list; `99a4a31b` adds `--engine` (default unchanged) and `resolution_degrees_built`; `b42e8494` adds `--arms`, `--no-degrees`, root digests and `compare`. No build changed the computation of a part it did not target. Driver logs: `run.log`, `run_eg3.log`, `run_wu2.log`, `run_frob4.log`.

Workloads

Koblitz curves `y^2 + xy = x^3 + a x^2 + 1` from `KoblitzCurve::new(a, n)`: K0/F2^9 `icv1-f2m9-t5-81e744be` (r = 127), K1/F2^15 `icv1-f2m15-t275-b7f03703` (r = 211), K0/F2^15 `icv1-f2m15-tm275-2d22ff5d` (r = 751), K1/F2^17 `icv1-f2m17-tm101-00378d4e` (r = 65 587), K0/F2^23 `icv1-f2m23-t5197-69e76b73` (r = 2 095 853), K1/F2^23 `icv1-f2m23-tm5197-1f85e9e1` (r = 4 196 903); targets uniform in the prime-order subgroup. Wu part also `icv1-f2m15-t5-d9fb183e` and `icv1-f2m17-t209-0eba1a1e` (a = 0, b != 1), targets uniform affine points. Factor bases: `eg`/`frob` use the divisor-of-`x^n - 1` subspace of `koblitz_symmetrised` (dimension ell in the JSON); `wu` uses `<1, z, ..., z^(ell-1)>`; `lucas` uses `<1, ..., z^(l-1)>` with `n = 3l` and random target abscissae.

Recorded values

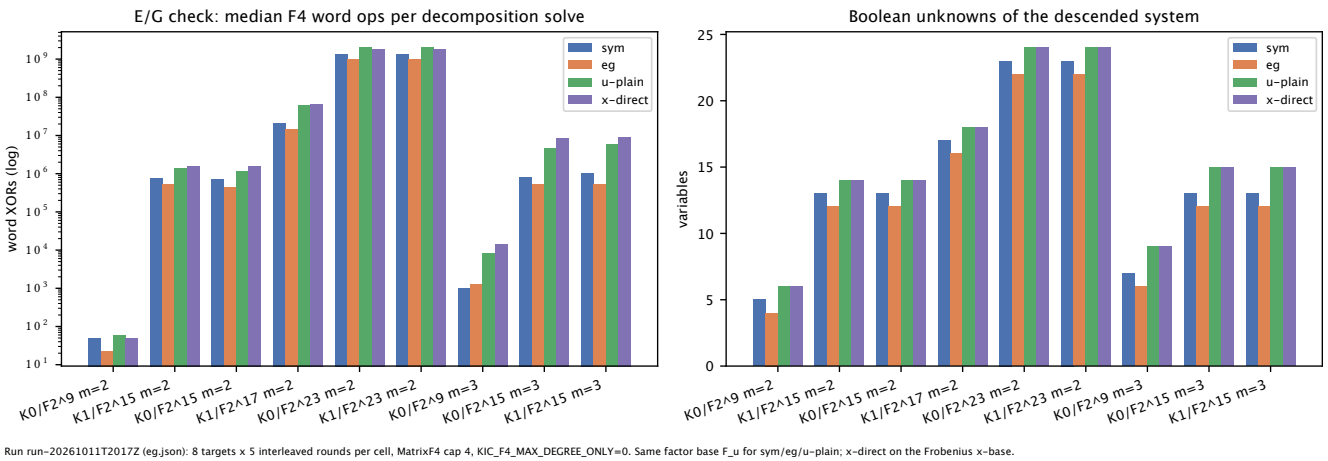


Figure 1: E/G check, `eg.json.partial`: median word XORs per solve and Boolean unknowns per arm and cell (8 targets x 5 rounds; MatrixF4 cap 4; `KIC_F4_MAX_DEGREE_ONLY=0`).

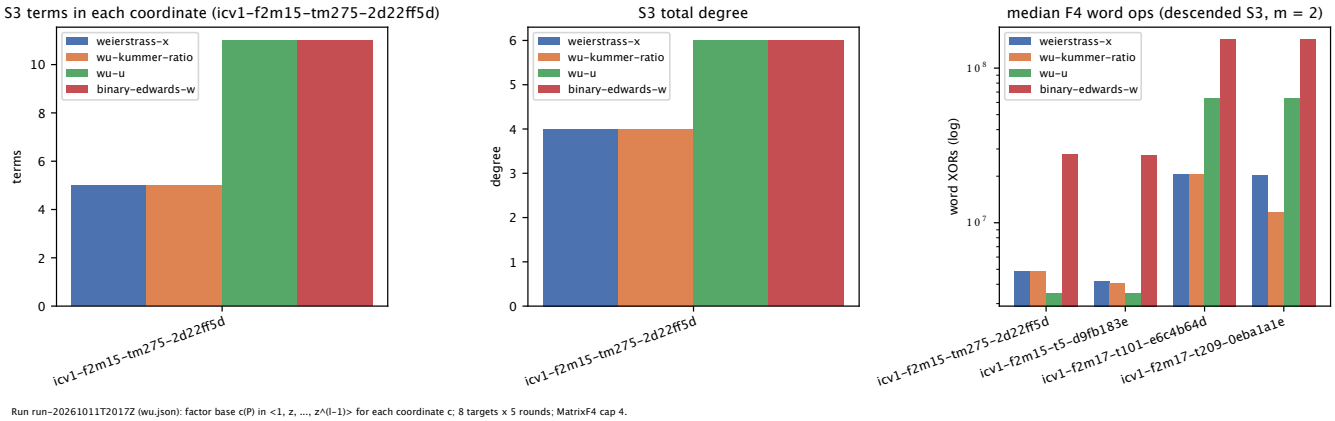


Figure 2: Wu part, `wu.json`: S3 terms and total degree per coordinate, and median word XORs per curve (8 targets x 5 rounds; MatrixF4 cap 4).

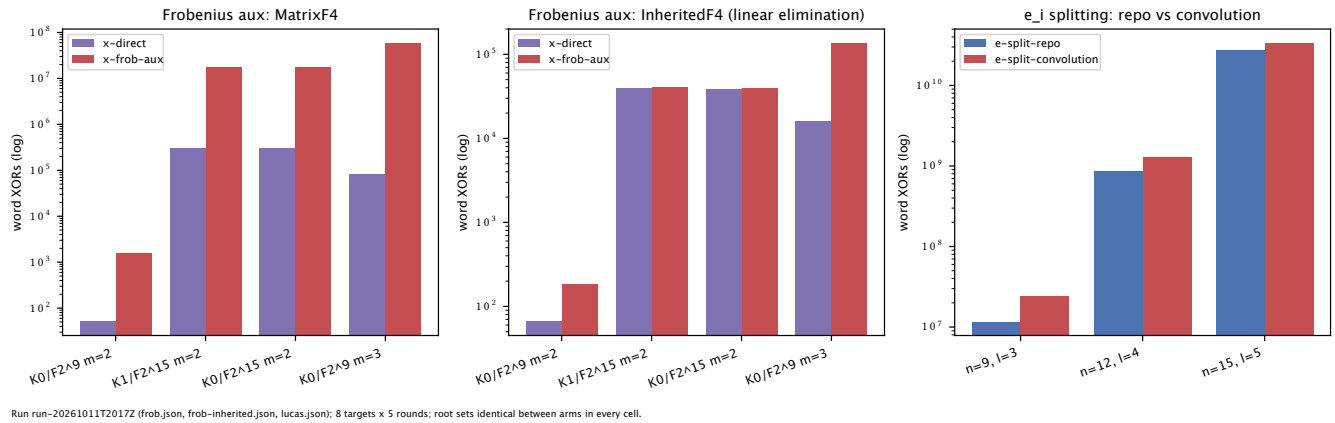


Figure 3: `frob.json`, `frob-inherited.json`, `lucas.json`: median word XORs per arm; root sets identical between arms in every cell.

Gates recorded

- `eg`: `w`-root sets of `sym` (parity dropped), `eg` and `u-plain` (image under AS) identical on all 72 targets; `sym` verdicts equal exhaustive enumeration on all 72; root sets stable across rounds; `b''` of `E/<T>` read and checked on 32 relations = 1 on every cell.
- `wu`: every model polynomial vanished on 32/32 random relations; every descended root set equal to brute force over `V x V` on 32/32 targets per model; Wu equation on 64/64 sampled points per curve; binary Edwards curve equation on 64/64 points and `w` invariant under `T` on 16/16.
- `frob`: projected root sets identical on 32/32 targets, both engines.
- `lucas`: projected root sets equal brute force over `V^3` on 24/24 targets, both splits.
- `cert`: certificates IDC1hc7b594938b50f160 (S3) and IDC1hcf83176ae9a93715 (S4) issue and check (32 points each).

Missing data: K1/F2^17 m=3 in the `eg` part (stopped). No other requested cell is missing.